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Stereo Vision System for Indoor Robot Navigation and Obstacle Avoidance

ITA0509-Computer Vision

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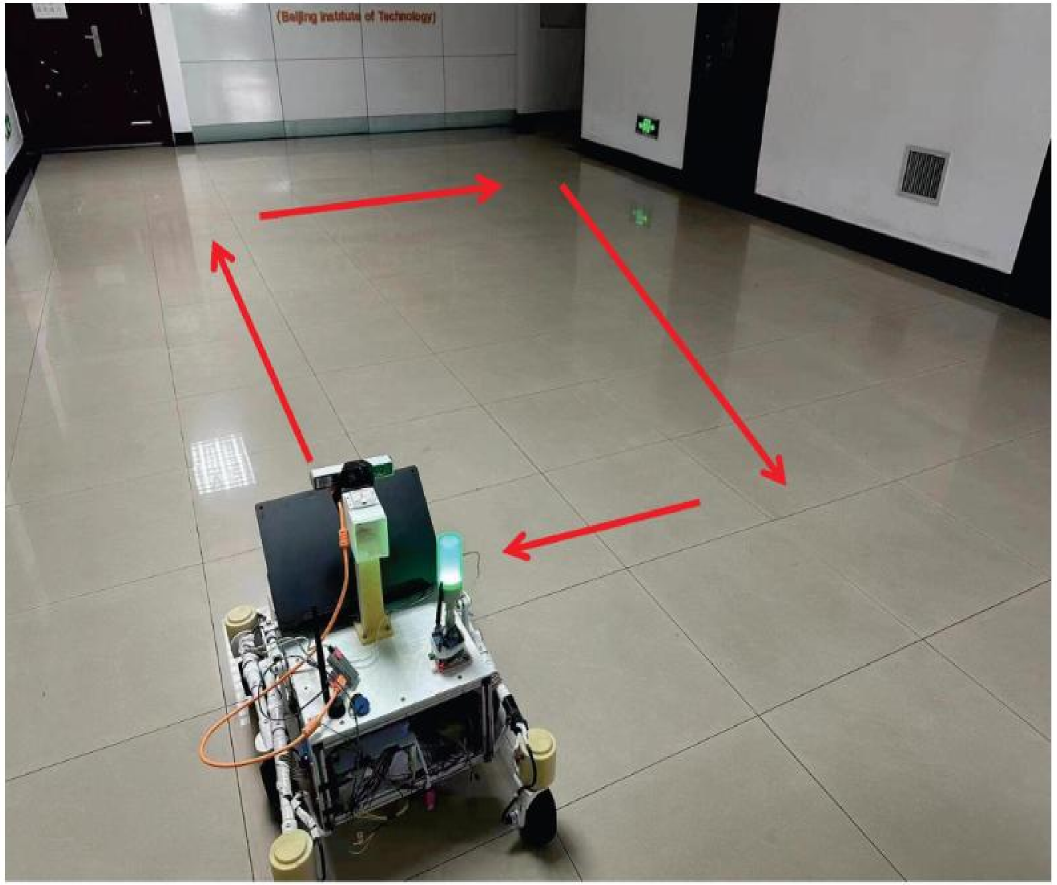
INTRODUCTION

- Indoor robots need to understand their surroundings for safe navigation.
- A single camera cannot accurately estimate the distance of objects.
- Stereo vision uses two cameras to estimate depth from two different viewpoints.
- The proposed system provides a software-based solution for indoor robot navigation.
- The system is developed and tested using simulation and datasets without physical hardware.



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OBJECTIVES

- To capture stereo images and calibrate the cameras.
- To estimate depth from stereo image pairs.
- To construct a 3D map of the indoor environment.
- To detect and classify obstacles using YOLO.
- To estimate the distance of obstacles from the robot.
- To create an occupancy grid of free and occupied spaces.
- To plan a safe and collision-free path using A/RRT.*
- To continuously replan the path when new obstacles appear.
- To control the robot for safe indoor navigation.
- To validate the complete navigation system through simulation.
- These expanded points are based on the modules and methodology in your uploaded

Problem Statement

- Indoor robots require reliable perception to understand the distance and position of surrounding objects.
- Conventional single-camera systems have difficulty obtaining direct depth information.
- Obstacles may be detected without accurately knowing their distance from the robot.
- Poor depth information can lead to unsafe navigation and collisions.
- LiDAR and dedicated depth cameras can provide depth information but may increase system cost.
- Therefore, there is a need for a **low-cost stereo vision-based system** that can estimate depth, detect obstacles and support safe indoor robot navigation.

Literature Review

S.No.	Title	Author(s)	Year	Problem Solved	Advantage	Disadvantage
1	Stereo Vision Based Obstacle Detection for Mobile Robots	B. Tivitmahaisoon, P. Prasant, et al.	2023	Accurate detection of obstacles in indoor environments using stereo vision for safe navigation.	Provides dense depth information and accurate obstacle localization.	Sensitive to lighting variations and textureless surfaces.
2	Real-Time Stereo Vision System for Indoor Mobile Robot Navigation	A. Hernandez, L. Kanade, et al.	2022	Real-time depth estimation and navigation in dynamic indoor environments.	Real-time performance with good depth accuracy and robustness.	High computational requirement for real-time processing.
3	Depth Map Generation Using Stereo Vision for Obstacle Avoidance	R. Gupta, M. Johnson, et al.	2023	Generation of reliable depth maps from stereo images to detect obstacles and plan safe path.	Improves obstacle detection accuracy using dense depth maps.	Depth estimation errors increase in low-texture regions.
4	Stereo Vision and SLAM Integration for Indoor Robot Navigation	H. Zhou, S. Li, et al.	2024	Combining stereo vision with SLAM for mapping and localization in indoor environments.	Enhances mapping accuracy and localization stability.	Cumulative drift may occur in long-term operations.
5	Obstacle Detection and Avoidance Using Stereo Vision in Mobile Robots	P. Singh, V. Sharma, et al.	2022	Detecting and avoiding static and moving obstacles using stereo vision system.	Effective in both static and dynamic obstacle environments.	Performance decreases in highly dynamic cluttered scenes.
6	Deep Learning Based Stereo Vision for Indoor Navigation	Y. Chen, J. Zhao, et al.	2024	Using deep learning for disparity estimation and obstacle detection in indoor scenarios.	Better generalization and robust performance in complex environments.	Requires large dataset and high-end GPU for training.
7	Energy Efficient Stereo Vision System for Indoor Robots	S. Park, T. Kim, et al.	2023	Designing an energy-efficient stereo vision pipeline for longer operation in indoor robots.	Lower power consumption with optimized processing pipeline.	Slight compromise in depth accuracy compared to full-resolution systems.

Proposed System

- The proposed system is a **stereo vision-based indoor robot navigation framework**.
- Two cameras capture synchronized left and right images of the environment.
- Camera calibration and rectification prepare the images for stereo matching.
- **SGBM** is used to calculate disparity between corresponding image points.
- Disparity information is converted into real-world depth.
- The depth information is used to identify obstacles and free space.
- **YOLO** is used for obstacle detection and classification.
- An occupancy grid is generated from the environment information.
- *A/RRT** is used for safe path planning.
- ROS/ROS2 integrates perception, planning and robot control.
- The complete system is evaluated through simulation.

Methodology

Stereo Cameras

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Camera Calibration & Rectification

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SGBM Stereo Matching

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Disparity & Depth Estimation

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3D Environment Mapping

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YOLO Obstacle Detection

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Occupancy Grid

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A/RRT Path Planning*

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Robot Control & Obstacle Avoidance

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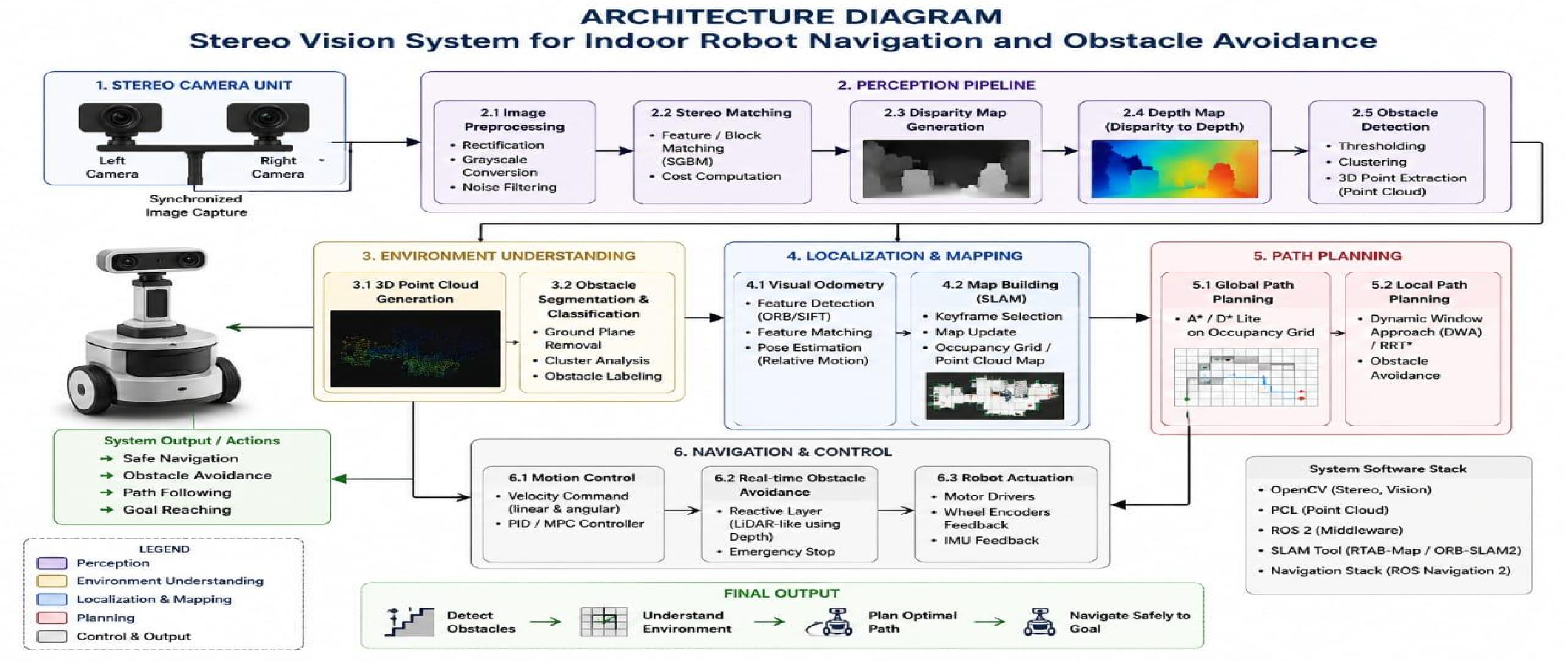
Safe Indoor Navigation

System Architecture



Module 1 :Stereo Image Acquisition & Camera Calibration

- Capture synchronized images using left and right stereo cameras.
- Obtain stereo images from a simulated environment or benchmark datasets.
- Determine camera intrinsic parameters such as focal length and optical center.
- Determine extrinsic parameters such as camera orientation and baseline.
- Use a checkerboard pattern for camera calibration.
- Correct lens distortion in both camera images.
- Rectify the left and right images so that corresponding points lie on the same scan line.
- Validate calibration accuracy using reprojection error.

MODULE 2: Depth Estimation & 3D Mapping

- Use the rectified left and right images as input for stereo matching.
- Apply **Semi-Global Block Matching (SGBM)** to calculate pixel disparity.
- Higher disparity indicates a closer object, while lower disparity indicates a farther object.
- Convert disparity into depth using camera focal length and baseline.
- Generate a dense depth map of the indoor environment.
- Convert depth information into a 3D point cloud.
- Apply filtering to reduce noise and incorrect stereo matches.
- Provide the resulting depth information to obstacle detection and navigation modules.

MODULE 3: Obstacle Detection & Classification

- Analyze the generated depth map to identify objects in the robot's environment.
- Extract regions containing potential obstacles.
- Use **YOLO** to detect and classify objects such as people, chairs, boxes and walls.
- Combine object detection results with depth information.
- Estimate the distance of each detected obstacle from the robot.
- Separate the environment into **free space and occupied space**.
- Generate an occupancy grid for navigation.
- Remove low-confidence detections to improve reliability.

MODULE 4: Path Planning & Navigation

- Use the occupancy grid as the representation of the robot's environment.
- Identify the robot's current position and destination.
- Use **A*** for efficient shortest-path planning in the grid.
- Use **RRT** when navigation involves more complex layouts.
- Generate a collision-free path around detected obstacles.
- Continuously update the path when the environment changes.
- Smooth the generated path into suitable navigation waypoints.
- Send the final path to the robot control module.

MODULE 5: Robot Control & Obstacle Avoidance

- Convert planned path waypoints into robot movement commands.
- Generate linear and angular velocity commands.
- Integrate perception, planning and control using **ROS/ROS2**.
- Continuously receive camera, depth and odometry information.
- Monitor the robot's surroundings while it is moving.
- Adjust the robot's velocity when a new obstacle is detected.
- Replan the path when the existing path becomes unsafe.
- Validate safe navigation through simulation.

Tools and Technologies

- **Python / C++** – System implementation
- **OpenCV** – Camera calibration, stereo processing and SGBM
- **YOLO** – Obstacle detection and classification
- **PyTorch / TensorFlow** – Deep learning model development
- **Gazebo / Webots** – Indoor robot simulation
- **ROS / ROS2** – Robot communication and system integration
- **KITTI Dataset** – Stereo vision benchmarking
- *A / RRT** – Path planning
- **RViz** – Robot and navigation visualization

Results and Analysis

- Stereo matching using SGBM generates disparity maps from left and right images.
- The disparity information provides depth information about the indoor environment.
- The generated depth map helps identify the location of nearby obstacles.
- YOLO detects and classifies important objects in the robot's environment.
- Depth information allows the system to estimate obstacle distance.
- The occupancy grid represents safe and occupied regions.
- A*/RRT generates collision-free navigation paths.
- The integrated pipeline enables the simulated robot to navigate while avoiding obstacles.
- The system is currently evaluated using simulation and datasets rather than physical hardware.

Advantages of the Proposed System

- Provides depth perception using a pair of cameras.
- Offers a comparatively **low-cost alternative to LiDAR or dedicated depth sensors**.
- Combines perception, obstacle detection and navigation in one framework.
- Supports automated obstacle detection and classification.
- Enables collision-free path planning.
- Uses simulation for safe testing and development.
- Modular architecture allows individual components to be improved independently.
- ROS/ROS2 integration supports future deployment on physical robots.
- Can be extended to different indoor environments.

Challenges / Limitations

- Stereo depth accuracy depends on camera calibration.
- Poor lighting can affect stereo correspondence.
- Textureless surfaces may produce inaccurate disparity.
- Occlusions can create missing or incorrect depth values.
- Real-time stereo processing can require significant computational resources.
- Simulation may not completely represent real-world sensor noise.
- YOLO performance depends on the quality and diversity of the training dataset.
- The proposed system has not yet been validated using a physical robot.

Future Enhancements

- Deploy the system on a real indoor mobile robot.
- Integrate a physical stereo camera system.
- Improve depth estimation under low-light conditions.
- Use a larger and more diverse obstacle dataset.
- Improve real-time performance on embedded hardware.
- Add detection and tracking of moving obstacles.
- Integrate dynamic path planning for changing environments.
- Improve 3D mapping accuracy.
- Add autonomous goal selection and navigation.
- Evaluate the system in real indoor environments such as laboratories, offices and corridors.

Conclusion

- A stereo vision-based system was designed for **indoor robot navigation and obstacle avoidance**.
- Stereo cameras provide the visual information required for depth estimation.
- Camera calibration and SGBM-based stereo matching generate disparity and depth maps.
- YOLO supports obstacle detection and classification.
- The occupancy grid provides environmental information for navigation.
- A*/RRT generates safe and collision-free paths.
- ROS/ROS2 integrates perception, planning and robot control.
- The complete pipeline has been evaluated using simulation and datasets.
- The proposed system provides a foundation for future real-world stereo vision-based autonomous robots.

References

1. Umam, F., Fuad, M., Suwarno, I., Ma'arif, A., and Caesarendra, W., "Obstacle Avoidance Based on Stereo Vision Navigation System for Omni-directional Robot," *Journal of Robotics and Control*, 2023.
2. Fan, X., Jahani Amiri, A., Fidan, B., and Jeon, S., "Unsupervised Stereo Matching with Surface Normal Assistance for Indoor Depth Estimation," *Sensors*, 2023.
3. Salehi, E., Aghagolzadeh, A., and Hosseini, R., "Stereo-RIVO: Stereo-Robust Indirect Visual Odometry," *Journal of Intelligent & Robotic Systems*, 2024.
4. Nguyen, C.-H., Vu, Q.-A., Phung Cong, K.-K., and Dang, T.-V., "Optimal Obstacle Avoidance Strategy Using Deep Reinforcement Learning Based on Stereo Camera," *MM Science Journal*, 2024.
5. Jiang, H., Lou, Z., Ding, L., Xu, R., Tan, M., Jiang, W., and Huang, R., "DEFOM-Stereo: Depth Foundation Model Based Stereo Matching," *CVPR*, 2025.

THANK YOU